

KWASI GYIMAH-ASANTE

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EXPERIENCE

Fordham University – Department of Computer Science

May 2026 – Present

Undergraduate Research Assistant

New York, NY

- Implemented a **value iteration** solver in **Python** to compute the exact optimal policy for entanglement scheduling on a quantum switch, formulating the scheduler as an **MDP** with fidelity-decay constraints and actions over swap, distill, and wait decisions
- Benchmarking the solver's provable optimum against a **MaskablePPO** (Stable Baselines 3) learned policy across fidelity thresholds, establishing a ground-truth baseline to evaluate RL policy quality, under Prof. Xiaolan Zhang

PROJECTS

Secure REST API | *Go, PostgreSQL, sqlc, chi, JWT, Docker*

[GitHub](#)

- Launched a production-ready **REST API** in **Go** featuring **JWT** authentication (**HS256**, 15-min access / 7-day refresh tokens) and **role-based access control** middleware enforcing per-route permission checks, structured across a clean layered architecture (handler → middleware → repository → DB)
- Implemented per-IP **token-bucket rate limiting** (10 req/s, burst 20) with automatic eviction of inactive clients after 3 minutes, and **bcrypt** password hashing at cost 14 for secure credential storage
- Automated schema provisioning by integrating **sqlc** for type-safe SQL query generation and applying migrations automatically on boot via **Docker Compose**, reducing local environment setup to a single command

Customer Query Routing Agent | *Python, Mistral 3B, Actian VectorAI, Docker*

[GitHub](#)

- Architected a multi-stage **agentic pipeline** with isolated **Triage**, **Resolution**, and **Escalation** service layers, routing queries against 4 **VectorAI** knowledge collections via **cosine similarity** ranking for autonomous resolution
- Improved retrieval precision by searching across FAQs, policy docs, resolved tickets, and a live **agent memory** collection that indexes runtime resolutions — surfacing ranked context before each **LLM** call; TODO: add benchmark figure here
- Ran inference fully locally using **Mistral 3B** via **llama-cpp-python**, eliminating external API dependency and enabling a fully self-contained deployment via **Docker**

Codebase Q&A Agent | *Python, Claude API, pgvector, Slack Bolt*

[Live Demo](#) | [GitHub](#)

- Built an **agentic RAG** pipeline where **Claude** autonomously issues up to **6 self-refined tool calls** per query via **tool use**, producing grounded answers with file/line citations
- Indexed **944 files (3,241 chunks)** in **31 seconds** by engineering an ingestion pipeline that clones **GitHub** repositories, chunks source files into overlapping windows, embeds them with **Voyage AI's** code-specialized embedding model, and stores as **1024-dim vectors** in **PostgreSQL** using **pgvector** for cosine similarity search
- Built **/ask** and **/ingest Slack Bolt** commands enabling real-time codebase Q&A and on-demand repository indexing, containerized with **Docker** for reproducible deployment

EDUCATION

Fordham University

August 2024 – May 2028

B.Sc. in Mathematics & Computer Science

- Relevant Coursework:** Computer Algorithms, Data Structures, Discrete Mathematics, Calculus I, II, III

CompTIA Security+

August 2025

Security Concepts, Threats & Mitigations, Security Architecture, Security Operations

TECHNICAL SKILLS

Languages: Python, Go, TypeScript/JavaScript, SQL (PostgreSQL)

Backend & Infrastructure: FastAPI, Node.js, PostgreSQL, Docker, OAuth 2.0, JWT, REST API Design, Rate Limiting

AI & ML: LLM Integration, RAG Pipelines, AI Agent Design & Tool Use, Embeddings & Vector Databases, Prompt & Context Engineering, Model Evaluation, Reinforcement Learning

Frameworks & Libraries: chi, HuggingFace, LangChain, OpenAI/Anthropic SDKs, Pydantic, NumPy, sqlc

Tools & Platforms: Git, GitHub Actions, Railway, AWS, Vercel, Supabase, Linux